

MyHealthReport

Version 1 Framework

22 August 2026

Agenda

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- 2 System Architecture
- 3 Module 1–3: Profile, Intake, Validation
- 4 Module 4–7: Ranges, Domains, Severity
- 5 Module 8–10: Patterns & Questions
- 6 Module 11–14: Context, Safety, Priority
- 7 Module 15–18: Recommendations & Explanation
- 8 Report, Tracking & Data Model
- 9 AI Role, Scope & Roadmap
- 10 Identity & Master Algorithm

1. Purpose of Version 1

MyHealthReport Version 1 is a **structured health interpretation and wellness-navigation platform**.

It helps a user to:

- 1 Understand their basic health profile
- 2 Upload and structure a blood test report
- 3 Identify out-of-range lab values
- 4 Understand patterns between results
- 5 Answer only relevant questions
- 6 Route findings: wellness vs. medical review
- 7 Receive a short, prioritised action plan
- 8 Understand the reasoning behind it
- 9 Track results over time

Not a diagnostic tool

Version 1 should **not** attempt to diagnose disease independently.

Data → Abnormality → Pattern → Context → Safety
→ Priority → Action → Explanation → Follow-up

2. Core Design Principles

1. Start simple

Registration collects only what's useful across almost every analysis; detailed questions come later, triggered by findings.

2. Never interpret one number in isolation

Result + Reference Range + Related Markers + Patient Profile + Symptoms + History + Trend

3. Safety overrides wellness advice

A clinically significant finding triggers medical assessment first; wellness advice becomes secondary.

2. Core Design Principles (cont.)

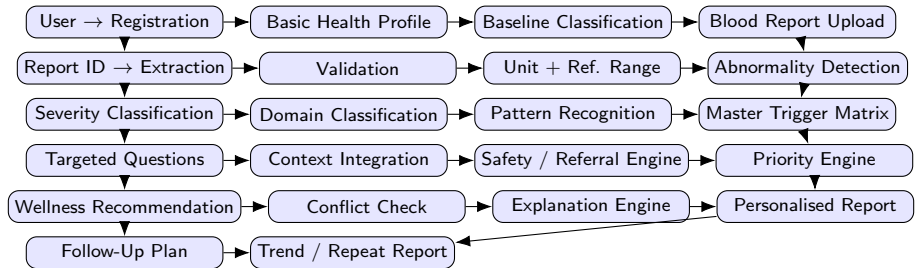
4. Explain the reasoning

Every recommendation answers: **What did we find? Why might it matter? What can you do? Why should that help?**

5. Prioritise instead of overwhelming

Twenty abnormal values should not produce twenty recommendations — group related abnormalities and surface the upstream priorities.

3. Version 1 System Architecture



4. Module 1: Basic Health Profile

Objective: enough baseline information to interpret future blood tests without a burdensome registration.

- Age / date of birth
- Biological sex
- Height, weight, waist circumference
- Blood pressure, if known
- Smoking status, alcohol use
- General physical activity
- Known major medical conditions
- Regular medication; pregnancy status

Automatically calculated

$$BMI = \frac{Weight (kg)}{Height (m)^2}$$

then classify weight status.

5. Module 2: Blood Report Intake

Objective: convert a laboratory report into structured data.

Step 1 — Identify Report: lab name, report date, patient identifiers, units, reference ranges.

Step 2 — Extract Results (structured record):

Field	Example
Test	ALT
Result	65
Unit	U/L
Reference range	0–40
Lab flag	High
Report date / source	22 Aug 2026 / Laboratory A

6. Module 3: Data Validation

Before interpretation, the system checks:

- Was the value extracted correctly?
- Does the unit make sense? Is the reference range available?
- Is the test name recognised? Is the result biologically plausible?
- Are there duplicates? Is a previous result available?

If confidence is low: user confirmation required

"We detected potassium 8.4 mmol/L. Please confirm that this value was read correctly from your report."

This prevents extraction errors from becoming health advice.

7. Module 4: Reference Range Architecture

Every laboratory marker carries **three layers**:

Layer A — Laboratory Reference Interval

Taken directly from the uploaded report, preserved exactly (e.g. ALT: 0–40 U/L).

Layer B — Clinical Decision Threshold

Recognised guideline thresholds, which may differ from the lab's population range.

Layer C — Platform Interpretation Rule

$$\text{Relative Elevation} = \frac{\text{Result}}{\text{Upper Limit of Normal}}$$

e.g. ALT 80 / ULN 40 = 2× ULN — comparable across labs.

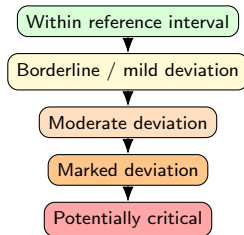
8. Module 5: Laboratory Domain Classification

- **A. Full Blood Count** — Hb, RBC, Hct, MCV, MCH, MCHC, RDW, WBC, differential, platelets
- **B. Glucose / Metabolic** — fasting glucose, HbA1c
- **C. Lipid Profile** — total cholesterol, LDL-C, HDL-C, triglycerides, non-HDL
- **D. Liver** — ALT, AST, ALP, GGT, bilirubin, albumin, total protein
- **E. Kidney** — creatinine, eGFR, urea, sodium, potassium
- **F. Uric Acid**
- **G. Thyroid** — TSH, free T4, free T3
- **H. Iron / Nutritional** — ferritin, iron, transferrin, TIBC, B12, folate, vitamin D

Version 1 focuses on common panels rather than every possible test.

9. Module 6: Abnormality Detection

Basic status: Normal / Low / High



Thresholds are marker-specific — not all “slightly outside range” values carry equal importance.

10. Module 7: Severity Framework — Six Action Levels

Level	Meaning
0	Within expected range — no major concern
1	Optimisation opportunity — not abnormal, but room to improve
2	Mild abnormality — lifestyle review, monitoring, follow-up
3	Clinically relevant — may warrant clinician assessment
4	High-risk finding — medical assessment prioritised
5	Potential critical finding — urgent safety pathway activated

11. Module 8: Pattern Recognition Engine

The system looks for **combinations**, not isolated values.

Metabolic Pattern

BMI↑, waist↑, triglycerides↑, HDL↓, glucose/HbA1c↑, BP↑, ALT↑ ⇒ *Metabolic-risk pattern* (not a diagnosis)

Microcytic Pattern

Hb↓, MCV↓, MCH↓ ⇒ ferritin/iron studies, blood loss & dietary history, referral logic

Liver Pattern

ALT/AST-predominant vs. ALP/GGT-predominant ⇒ different question sets

12. Module 9: Master Trigger Matrix

LAB FINDING → PATTERN → TRIGGER → TARGETED QUESTION SET

Trigger	Question
ALT elevated	Alcohol intake / Medication / Supplements
ALT elevated	Known fatty liver / Recent heavy exercise
ALT + metabolic markers abnormal	Weight change
GGT elevated	Alcohol context
Significant abnormality	Relevant symptoms

The patient answers only the questions that matter to their results.

13. Module 10: Targeted Question Engine

- **Symptoms** — fatigue, dizziness, chest discomfort, breathlessness, jaundice, bleeding, severe weakness
- **Medical History** — diabetes, hypertension, kidney/liver/thyroid/cardiovascular disease
- **Medication** — regular, recent, supplements, herbal products
- **Diet** — targeted, e.g. triglycerides high → sugary drinks, refined carbs, alcohol, overeating
- **Exercise** — frequency, duration, intensity, resistance work, sedentary time
- **Sleep** — duration, quality, irregularity, sleep-disordered breathing
- **Stress** — low / moderate / high + major stressors

14. Module 11: Context Integration Engine

LAB RESULTS + PATIENT PROFILE + MEDICAL HISTORY
+ MEDICATIONS + SYMPTOMS + LIFESTYLE + PREVIOUS RESULTS

This creates the health context used by later engines.

15. Module 12: Safety and Referral Engine

Main question: Is this situation appropriate for routine wellness guidance?

If YES

Routine Wellness Pathway — no major safety concern identified.

If NO — escalate

Medical Review Recommended → Prompt Medical Assessment → Urgent Assessment

Exact clinical thresholds are built separately in the **Safety Matrix** and medically validated before implementation.

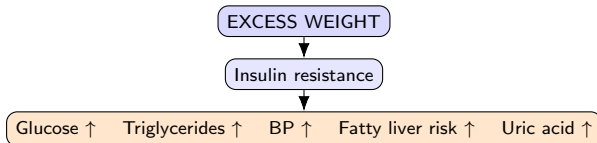
16. Module 13: Priority Engine

Example patient: obesity, elevated BP, high triglycerides, elevated glucose, mild ALT elevation, high uric acid — instead of six separate recommendations:

- ① **Priority 1:** Metabolic health / weight management
- ② **Priority 2:** Blood pressure
- ③ **Priority 3:** Cardiovascular risk
- ④ **Priority 4:** Liver monitoring

This makes the action plan easier to follow.

17. Module 14: Root-Factor Mapping



"Weight management is prioritised because it may positively influence several current metabolic findings at the same time."

18. Module 15: Wellness Recommendation Engine

Version 1 recommendations focus on:

- Nutrition
- Physical activity
- Weight management
- Sleep
- Stress management
- Smoking cessation / alcohol moderation
- Hydration where appropriate
- General preventive health behaviour

Advice should be proportional to the evidence available.

19. Recommendation Structure

TRIGGER → HEALTH ISSUE → RECOMMENDATION → WHY → EXPECTED BENEFIT → PRECAUTION → FOLLOW-UP

Example

Trigger: Elevated triglycerides + excess body weight

Recommendation: Reduce refined carbs and sugary beverages; gradual weight reduction

Why: Excess energy / refined-carb intake contributes to elevated triglycerides

Expected Benefit: Improved triglycerides and broader metabolic health

Precaution: May need modification per medical conditions, medication, nutrition status

Follow-Up: Repeat relevant measurements per clinical plan

20. Module 16: Recommendation Conflict Resolver

All recommendations pass a safety filter before being shown.

Recommendation	Conflict	Action
Increase exercise intensity	Severe anaemia / cardiac symptoms	Do not issue routine intensive advice
High-protein diet	Kidney impairment	Modify or suppress pending assessment
Aggressive weight loss	Pregnancy / underweight	Suppress recommendation

21. Module 17: Recommendation Priority System

Each recommendation is scored on:

- Clinical importance
- Modifiability
- Number of findings affected
- Safety
- Expected health benefit
- User feasibility

Target output

Approximately **3 main priorities**, with secondary recommendations underneath. Avoid ten equally-weighted actions.

22. Module 18: Explanation Engine

Every important recommendation offers:

- **What we found** — “Your triglyceride level is above the laboratory reference range.”
- **Why it may matter** — “Elevated triglycerides can be associated with metabolic and cardiovascular risk.”
- **What may contribute** — excess weight, dietary pattern, alcohol, glucose regulation, inactivity
- **What you can do** — specific actions
- **Why this may help** — mechanistic explanation
- **What happens next** — monitoring, repeat testing, or professional assessment

23. Module 19: Final User Report Structure

- 1 **Health Snapshot** — BMI, BP category, smoking status, activity level, baseline risks
- 2 **Laboratory Overview** — Within range / Needs attention / Higher priority / Medical review recommended
- 3 **Key Patterns** — e.g. metabolic-risk, iron-related, liver enzyme abnormality
- 4 **Your Top Priorities** — usually three
- 5 **Recommended Actions** — concrete, measurable
- 6 **Why These Actions Matter** — causal reasoning
- 7 **Medical Follow-Up** — separate from wellness advice
- 8 **Monitoring Plan** — weight, waist, BP, activity, labs, symptoms

24. Module 20: Longitudinal Tracking

Date	HbA1c
Jan	5.6
Apr	5.8
Aug	6.1

Instead of “HbA1c is elevated,” the system can say: “*Your HbA1c has shown a progressive upward trend across three measurements.*” **Trend categories:** Improving, Stable, Worsening, Fluctuating

25. Version 1 Database Concept

Core structured objects (not free text):

- Patient
- Laboratory Report
- Laboratory Result
- Health Domain
- Trigger
- Question
- Answer
- Pattern
- Safety Rule
- Recommendation
- Contraindication
- Priority
- Explanation
- Follow-Up

26. Master Rule Architecture

```
IF Finding X AND/OR Finding Y AND Patient Context Z  
THEN Activate Question Set A  
THEN Assess Safety Rule B  
THEN Identify Pattern C  
THEN Generate Recommendation D  
UNLESS Contraindication E  
THEN Assign Priority F AND Generate Explanation G  
AND Generate Follow-Up H
```

The standard structure for almost every clinical/wellness rule in the platform.

27–28. The Role of AI

AI **should** do

- Report extraction assistance
- Natural-language explanation
- Summarisation & personalised wording
- Identifying relevant rules
- Explaining relationships between findings
- Generating patient-friendly reports

AI should **not** independently control

- Emergency triggers, critical thresholds
- Contraindications, referral criteria
- Diagnostic claims
- Medication advice
- Clinically sensitive recommendations

Structured clinical rules determine what is allowed. AI determines how it is explained.

29. Version 1 Scope Boundary

Include

Basic profile, BMI, BP context, common blood tests, abnormality detection, severity classification, common pattern recognition, targeted questions, wellness recommendations, red-flag/referral logic, explanations, trend tracking, report generation

Delay to later versions

Supplement marketplace, automatic supplement prescribing, pharmacy sales integration, complex genetic interpretation, imaging interpretation, diagnosis engine, medication initiation/adjustment, complex specialist disease management, autonomous AI clinical decisions

30. Clinical Domains — Build Order

- ① **Phase 1 — Metabolic / cardiovascular:** BMI, BP, glucose, HbA1c, lipid profile
strong preventive-health foundation
- ② **Phase 2 — Liver + kidney**
- ③ **Phase 3 — Full blood count + iron**
- ④ **Phase 4 — Thyroid + uric acid + nutritional markers**

Do not attempt to build every laboratory module simultaneously.

31. The Four Master Matrices

A. Laboratory Interpretation Matrix

Marker, Reference logic, Severity, Related markers, Potential patterns

B. Trigger Question Matrix

Finding, Trigger, Question, Why, Possible responses

C. Safety / Referral Matrix

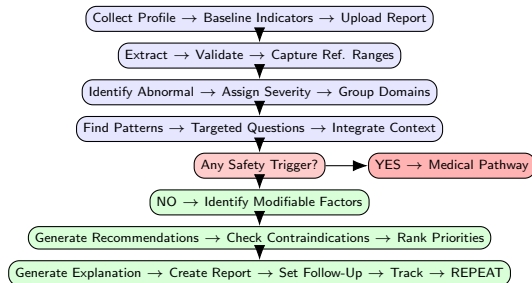
Finding, Threshold/combination, Risk level, Action, Referral urgency

D. Recommendation Matrix

Finding/pattern, Recommendation, Why, Contraindications, Priority, Follow-up

These four matrices are the **brain of MyHealthReport**.

32. Version 1 Decision Flow



33. The Identity of MyHealthReport

Not: “AI tells you what your blood test means.”

But: “MyHealthReport turns your blood results, health profile and lifestyle into a personalised health roadmap.”

Understand Prioritise Act Escalate Explain Track

34. The Central MyHealthReport Algorithm

Health Interpretation = Lab Findings + Patterns + Patient Context + Risk + Trend

Action Plan = Priority × Modifiability × Benefit × Safety

subject to

Safety Override > Wellness Recommendation

MyHealthReport = Understand + Prioritise + Act + Explain + Monitor

This is the Version 1 master architecture. The next layer is the **Master Clinical Logic System**: Laboratory Interpretation, Trigger, Safety, and Recommendation Matrices.

Thank You

MyHealthReport — Version 1 Framework